Practical: Active Directory Certificate Services (AD CS)

Aim

To install and configure **Active Directory Certificate Services (AD CS)** on Windows Server, create a Standalone Root Certification Authority (CA), issue a user certificate, configure client certificate authentication for a web application, and troubleshoot common AD CS configuration errors.

Requirements

- Windows Server VM
- Administrator account
- Active Directory Domain Services (if required for user management)
- IIS (Internet Information Services)
- Network connectivity
- Test user account: testuser

Task 1 – Install Active Directory Certificate Services (AD CS)

Objective

Install the AD CS role on the Windows Server and select the **Certification Authority** role service.

Procedure

Step 1: Open Server Manager

1. Log in to the Windows Server using an Administrator account.
2. Open **Server Manager**.
3. Click **Manage**.
4. Select **Add Roles and Features**.

Step 2: Add the AD CS Role

1. The **Add Roles and Features Wizard** will open.

2. Click **Next**.
3. Select **Role-based or feature-based installation**.
4. Click **Next**.
5. Select the appropriate server from the server pool.
6. Click **Next**.
7. On the **Server Roles** page, select:

Active Directory Certificate Services

8. When prompted, click **Add Features**.
9. Click **Next**.
10. Continue through the Features page and click **Next**.
11. Read the AD CS information and click **Next**.

Step 3: Select Role Services

On the **Role Services** page:

1. Select:

Certification Authority

2. Click **Next**.
3. Review the installation settings.
4. Click **Install**.
5. Wait for the installation to complete.

Screenshot

Take a screenshot of the **Role Services** page showing:

Certification Authority selected.

Task 2 – Configure a Standalone Root Certification Authority

Objective

Configure a new **Standalone Root CA** and set its Common Name to **CampusAppCA**.

Procedure

Step 1: Open AD CS Configuration

1. After the AD CS installation is completed, open **Server Manager**.
2. Click the **Notification Flag** in the upper-right corner.
3. Select:

Configure Active Directory Certificate Services on the destination server

Step 2: Select Server and Role Service

1. Select the appropriate server.
2. Select:

Certification Authority

3. Click **Next**.

Step 3: Select Setup Type

For **Certification Authority Setup Type**, select:

Standalone CA

Click **Next**.

Step 4: Select CA Type

For **CA Type**, select:

Root CA

Click **Next**.

Step 5: Configure Private Key

For **Private Key**, select:

Create a new private key

Click **Next**.

Step 6: Configure Cryptography

Leave the default cryptographic settings.

Click **Next**.

Step 7: Set Common Name

Enter the following Common Name:

CampusAppCA

Click **Next**.

Step 8: Set Validity Period

Accept the default validity period.

Click **Next**.

Step 9: Configure CA Database

Accept the default database and log locations unless your practical specifically requires different locations.

Click **Next**.

Step 10: Complete Configuration

1. Review the configuration summary.
2. Click **Configure**.
3. Wait until the configuration is completed successfully.
4. Click **Close**.

Expected Result

A Standalone Root Certification Authority named:

CampusAppCA

is successfully created.

Task 3 – Issue and Export a User Certificate

Objective

Create a test user account, issue a user certificate from the CA, and export the certificate as a .cer file.

Step 1: Create Test User

1. Open **Server Manager**.

2. Select **Tools → Active Directory Users and Computers**.
3. Navigate to the appropriate domain or Users container.
4. Create a new user.
5. Enter:

Username: testuser

6. Set a password for the account.
7. Complete the user creation wizard.

Step 2: Open Certification Authority

1. Open **Server Manager**.
2. Select:

Tools → Certification Authority

3. Expand the CA:

CampusAppCA

Step 3: Verify Certificate Template

If certificate templates are available, verify that a suitable **User** certificate template is enabled.

For a Standalone CA, certificate enrollment may differ from an Enterprise CA. If the User template is not available, use the certificate enrollment method supported by the configured CA.

Step 4: Request a Certificate

1. Open **MMC**.
2. Click **File → Add/Remove Snap-in**.
3. Select **Certificates**.
4. Click **Add**.

5. Select:

My user account

6. Click **Finish → OK**.
7. Expand:

Certificates → Current User → Personal

8. Right-click **Certificates**.
9. Select:

All Tasks → Request New Certificate

10. Follow the Certificate Enrollment Wizard.
11. Select the appropriate **User** certificate.
12. Complete the wizard.

Expected Result

A user certificate for testuser is issued and appears under:

Current User → Personal → Certificates

Step 5: Export the Certificate

1. Right-click the issued certificate.
2. Select:

All Tasks → Export

3. The Certificate Export Wizard will open.
4. Select:

No, do not export the private key

5. Click **Next**.
6. Select an appropriate certificate format such as **DER encoded binary X.509 (.CER)**.
7. Click **Next**.
8. Enter the file name:

testuser.cer

9. Select a suitable location.
10. Click **Finish**.

Expected Result

The certificate is successfully exported as:

testuser.cer

Task 4 – Configure Client Certificate Authentication for a Web Application

Objective

Simulate a secure login system similar to an online shopping website by requiring a client certificate to access a test web application.

Step 1: Install IIS

1. Open **Server Manager**.
2. Click **Manage → Add Roles and Features**.
3. Select **Web Server (IIS)**.
4. Install the required IIS role services.
5. Complete the installation.

Step 2: Create a Simple Web Page

1. Open:

C:\inetpub\wwwroot

2. Create a file named:

index.html

3. Add a simple webpage, for example:

<!DOCTYPE html>

```
<html>

<head>

  <title>Campus Secure Login</title>

</head>

<body>

  <h1>Campus Secure Login</h1>

  <p>Client certificate authentication is enabled.</p>

</body>

</html>
```

4. Save the file.

Step 3: Open IIS Manager

1. Open **Server Manager**.
2. Select:

Tools → Internet Information Services (IIS) Manager

3. Expand the server.
4. Expand **Sites**.
5. Select **Default Web Site**.

Step 4: Configure HTTPS/SSL

To use client certificates, the website should be accessed through HTTPS.

1. Configure an HTTPS binding for the website.
2. Select an appropriate server certificate.
3. Open **SSL Settings** for the website.
4. Enable:

Require SSL

5. Under client certificates, select:

Require

This forces the client to provide a certificate when connecting to the website.

Step 5: Configure Client Certificate Authentication

1. Open **Authentication** for the website.
2. Enable the appropriate client certificate authentication feature.
3. If required by the IIS configuration, configure **IIS Client Certificate Mapping Authentication**.
4. Map the certificate associated with testuser to the testuser account.

Note: The exact certificate-mapping options available depend on the Windows Server/IIS configuration. Certificate mapping should be configured carefully because merely enabling client certificates does not automatically map a certificate to a Windows user.

Step 6: Test the Secure Website

1. Open a browser from a client machine or the server.
2. Enter:

https://ServerName

3. The browser should request a client certificate.
4. Select the certificate issued to testuser.
5. Continue to the website.

Expected Result

The browser successfully connects to the HTTPS website using the client certificate.

This demonstrates how certificate-based authentication can be used as an additional security mechanism for a web application.

Task 5 – Troubleshoot AD CS Installation and Configuration Errors

Objective

Use a troubleshooting checklist to identify and fix common AD CS configuration problems.

AD CS Troubleshooting Checklist

Step 1: Verify Administrator Access

Confirm that you are logged in using an account with the required administrative privileges.

- Verify Administrator access.
- Confirm the server is accessible.

Step 2: Verify AD CS Installation

Open **Server Manager → Manage → Remove Roles and Features** or review the installed roles.

Confirm that:

Active Directory Certificate Services

is installed.

Step 3: Check Required Windows Services

Open:

Services → services.msc

Verify that the following services are running:

- **Remote Procedure Call (RPC)**
- **Windows Event Log**
- **DCOM Server Process Launcher**

If a required service is stopped, start it and test the AD CS configuration again.

Step 4: Check Event Viewer

Open:

Event Viewer

Check:

Windows Logs → Application

and

Windows Logs → System

Look for errors or warnings related to AD CS, certificates, services, permissions, or RPC communication.

Step 5: Verify CA Configuration

Open the **Certification Authority** console and verify:

- CA Type
- Private Key
- Common Name
- CA status

The configured CA should have the correct name:

CampusAppCA

Step 6: Restart AD CS Service

If required, restart the AD CS service.

Verify that the service starts successfully after the restart.

Step 7: Verify the CA Console

Open:

Server Manager → Tools → Certification Authority

Verify that:

CampusAppCA

appears in the Certification Authority console.

Intentional Misconfiguration

To demonstrate troubleshooting, intentionally select an incorrect CA type during configuration.

Incorrect Configuration

Select:

Subordinate CA

instead of:

Standalone CA → Root CA

A Subordinate CA requires an existing parent Certification Authority. Therefore, attempting to configure it without a suitable parent CA should result in a configuration problem or failure.

Fixing the Misconfiguration

1. Open the **Configure Active Directory Certificate Services** wizard.
2. Select the correct server.
3. Select:

Certification Authority

4. Select:

Standalone CA

5. Select:

Root CA

6. Select:

Create a new private key

7. Accept the default cryptographic settings.
8. Set the Common Name to:

CampusAppCA

9. Accept the default validity period.
10. Complete the configuration wizard.
11. Open the **Certification Authority** console.
12. Verify that **CampusAppCA** is displayed.

Final Verification

Perform the following checks after completing the practical:

- AD CS role is installed.
- Certification Authority role service is installed.
- CampusAppCA is configured as a Standalone Root CA.
- testuser account is created.
- User certificate is issued successfully.
- Certificate is exported as testuser.cer.
- IIS is installed.
- HTTPS is configured.
- Client certificate authentication is enabled.
- testuser certificate can be selected by the browser.
- The secure test webpage opens successfully.
- CampusAppCA appears in the Certification Authority console.